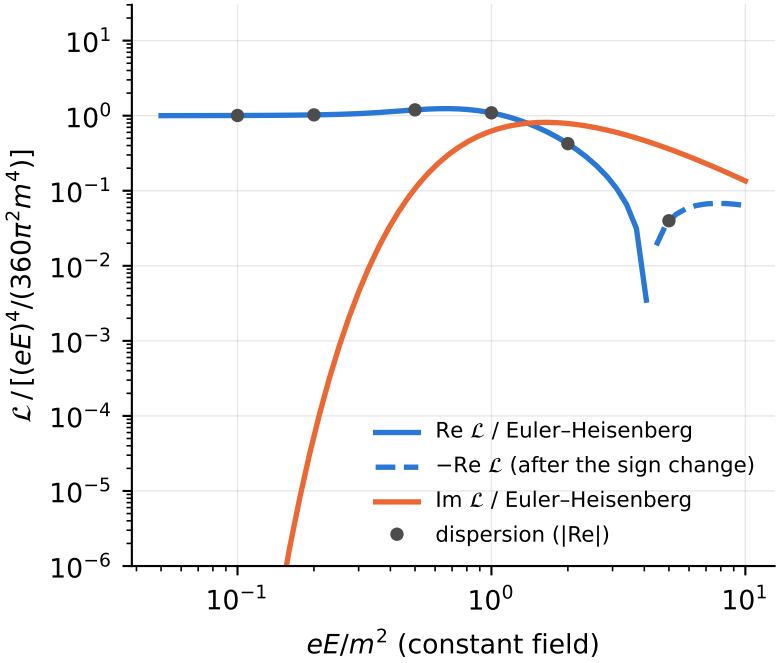
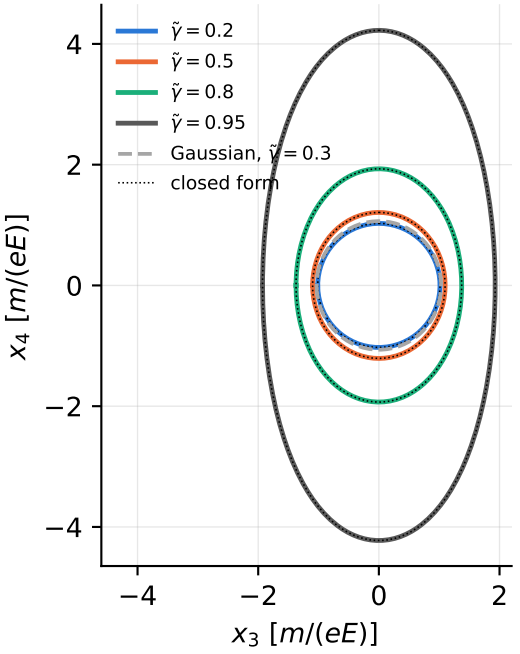
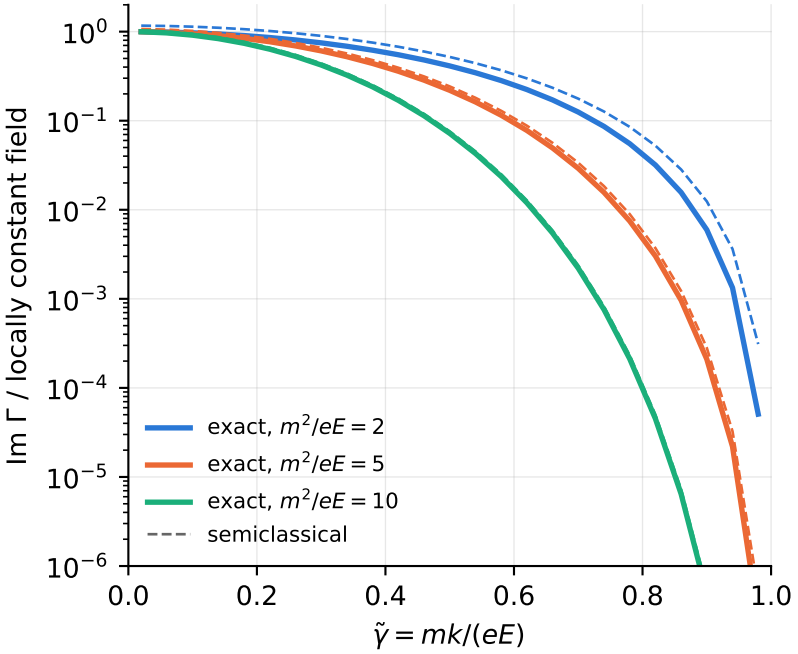

$\mathcal{L}/[(eE)^4/(360\pi^2 m^4)]$
eE/m^2 (constant field)
Re $\mathcal{L}$ / Euler–Heisenberg
−Re $\mathcal{L}$ (after the sign change)
Im $\mathcal{L}$ / Euler–Heisenberg
dispersion (|Re|)


$\tilde{\gamma} = 0.2$
$\tilde{\gamma} = 0.5$
$\tilde{\gamma} = 0.8$
$\tilde{\gamma} = 0.95$
Gaussian, $\tilde{\gamma} = 0.3$
closed form
x_4 $[m/(eE)]$
x_3 $[m/(eE)]$


Im Γ / locally constant field
exact, $m^2/eE = 2$
exact, $m^2/eE = 5$
exact, $m^2/eE = 10$
semiclassical
$\tilde{\gamma} = mk/(eE)$